

TorqFall — DIY Build Guide v1

0. What we build, what we buy

Everything the [TorqFall — Venue Scenarios & Build Plan v1](#) §4 table marks **DIY**, turned into a shopping list, a cut list and a step-by-step. TorqFall has no hard barricade to build in any scenario, and the driver stand is a rental everywhere — its DIY list is the gauntlet obstacle kit plus the pit. Prices are Sept-2026 India ballparks — check the live listing before ordering. Amazon links are **search links**, not fixed listings, so they keep working when a seller changes.

Item	Build time (2–3 people)	Est ₹	Section
Start/finish arches + LED — 2	Half a day	3,700–6,900	§2
Pendulum + sweeper bars (incl. 1 spare sweeper)	1 day	1,600–2,750	§3
Spinner gate (2 drives, bench-tested) + Ramp Scramble	1.5 days	5,950–11,900	§4
Soft barrier rail + bump rings + queue posts	1 day	9,300–14,200	§5
Collapse Gate wall stock + scoring jig	2 hours/event (jig one-time)	2,000–2,500 first event · 500–1,000/event after	§6
Pit + charging table	Buy + 1 hour setup	12,700–23,900	§7
Tools, if we own none	—	6,500–10,500	§1

Not in this guide (vendor): turf (lane + Tier-3 pen), driver stand, branding print, DJ/sound, food, the rented RC fleet + pit crew, and (Scenario C only) the weatherproof tent + portable AC — see the Venue Scenarios doc §4.

1. Tools + consumables — one shopping run

Buy once, reuse every event. Skip anything already in the garage.

Tools

Tool	Why	Est ₹	Where
Cordless drill/driver, 12–18 V, with bit set	Every screw in this guide	2,500–4,500	amazon.in
Jigsaw	Ramp core, slope cuts	2,500–4,500	amazon.in

Tool	Why	Est ₹	Where
Hand saw (PVC)	Cuts every pipe length in this kit	300–500	amazon.in
Measuring tape 5 m + carpenter's square	Marking the lane, gate spacing, pipe lengths	300–600	amazon.in
Utility knife + spare blades	Foam/noodle/cardboard cuts	150–300	amazon.in
Steel ruler + cutting mat (Collapse-Gate scoring jig)	Scores cardboard seams without cutting through	800–1,200	amazon.in
Sandpaper 80 + 120 grit	Every cut edge gets sanded — splinters snag a chassis	200–400	amazon.in
Hot-glue gun + sticks	Tacking LED strip	300–600	amazon.in
Basic multimeter	Bench-testing the spinner gearmotors before mounting	400–700	amazon.in
Safety glasses + dust masks	Ply dust, cardboard dust	300–500	amazon.in

Consumables (shared across every build below)

Item	Qty	Est ₹	Where
Cable ties 200 mm + 300 mm, black	200 total	500–800	amazon.in
Gaffer tape, black, 3 rolls	3	700–1,200	amazon.in
Empty sandbags (fill locally)	60	2,200–4,000	amazon.in
Sand	4 bags	400–700	Local hardware / construction supplier
Heavy-duty adhesive velcro, 25 mm	3 m	250–500	amazon.in

Local, not Amazon: 25 mm PVC pipe in 3 m lengths, 9 mm plywood, pine battens (timber yard, take §8's cut plan). Every Mumbai timber/pipe yard cuts to size for ₹20–40 a cut.

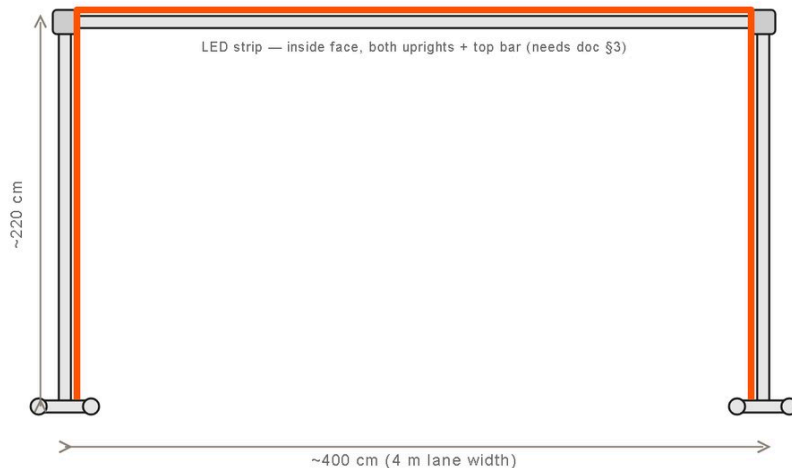
2. Start / finish arches

Start / finish arch — PVC + LED

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2 arches (start + finish) · 25 mm PVC frame spans the ~4 m lane width (rulebook §3) · 5 m USB LED strip each side (needs doc §3)

A · FRONT VIEW — spans the lane's 4 m width



PVC CUT LIST — per arch (25 mm pipe)

Uprights	2 x 220 cm
Top bar	1 x 400 cm (join 2 lengths)
Feet	2 x 30 cm
Elbows 90°	2
Tees (feet)	2
End caps	4
LED strip	2 x 5 m (uprights + bar)

TEST

Roll a car through at speed — the arch must not flex or shed the LED strip. Re-tape any loose run.

Dry-fit only — friction-fit PVC, no glue, comes apart for transport.
Feet run front-to-back (like the TorqBall goal) so a shoved car can't tip the arch; a sandbag sits on each foot on event day.
USB battery pack (per needs doc §3) gaffer-taped to one foot.
Arch height/width are a build choice — not dimensioned in the docs;
keep it clear of the overhead cam's sightline (rulebook §8).

Two arches, one each end of the lane, spanning the **~4 m lane width** (rulebook §3) — a build choice, since the docs don't dimension the arch itself. 25 mm PVC frame, USB LED strip on a power bank, no soldering, no mains near the lane.

Materials (for 2 arches)

Item	Qty	Est ₹	Where
25 mm PVC pipe, 3 m	3 lengths	550–850	Local hardware
25 mm PVC 90° elbows	4	100–200	amazon.in
25 mm PVC tees	4	100–200	amazon.in
25 mm PVC end caps	8	100–150	amazon.in
25 mm PVC straight coupler	2	80–150	amazon.in
USB 5 V LED strip, 5 m	4	1,200–2,400	amazon.in
Power bank, 10,000 mAh	2	1,600–3,000	amazon.in
Sandbags (from §1 stock)	4	—	—
Arches total		≈ 3,700–6,900	

Cut list (per arch)

- 25 mm PVC: 2 × 220 cm uprights, 1 × 400 cm top bar (join two ~200 cm lengths with the straight coupler), 2 × 30 cm feet

Steps

1. **Dry-fit** each arch: foot tee → upright → elbow → top bar (joined at centre with the coupler) → elbow → upright → foot tee. Friction-fit only — no glue, so it comes apart for transport.
2. Stand it at the lane's start or finish line; check it clears the ~4 m width with room to spare and sits clear of the overhead camera's sightline (rulebook §8).
3. **LED strip:** peel-and-stick along the inside face of both uprights and the top bar. Hot-glue every 30 cm — adhesive alone lets go on PVC.
4. Route the USB lead down one upright to the foot; gaffer-tape the power bank to the outside of that foot.
5. One sandbag on each foot — feet run front-to-back so a shoved car can't tip the arch.
6. **Test:** roll a car through at speed. If the arch flexes or sheds the LED strip, re-tape the run and re-check the foot joints.

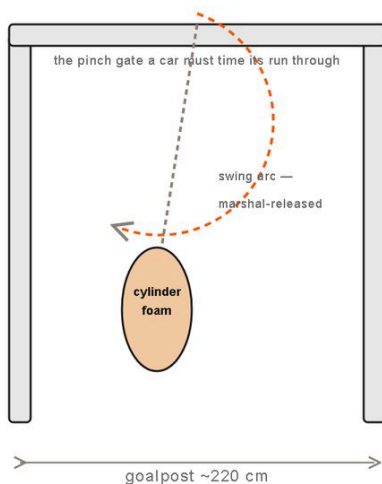
3. Pendulum + sweeper bars

Pendulum + sweeper bars

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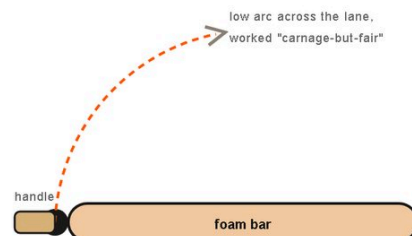
Foam only, low mass, at car height (rulebook §7) · marshal-worked, springs back <3 s between heats (rulebook §3)

A · PENDULUM — hangs off the goalpost frame over a pinch gate



Rope + foam cylinder, hung from the top bar.
Marshal releases it on a rhythm — mistime it,
get swatted back (Level Directory, Tier-2 #1).

B · SWEEPER BAR — foam bar + handle, marshal-swung



Make 2 + a 3rd spare — one live + one for
Twin Sweepers (instant swap), 3rd for
Hex-a-Fall which runs all 3 at once
(needs doc §3). Resets in the marshal's
hand — no mechanism, no delay.

The signature wipe-out obstacles. Foam only, low mass, at car height (rulebook §7) — the pendulum hangs off a goalpost frame over a pinch gate; the sweeper bar is marshal-swung across the open lane.

Materials

Item	Qty	Est ₹	Where
25 mm PVC pipe, 3 m (goalpost frame)	2 lengths	350–550	Local hardware
25 mm PVC elbows + tees	4 + 2	150–300	amazon.in
Foam cylinder (bundled pool noodles, wrapped)	1	300–500	amazon.in
Nylon rope, 5 m	1	150–250	amazon.in
Pool noodles (sweeper bar cores — 2 live + 1 spare)	3	450–750	amazon.in
Broom-handle / PVC handle stock	3	200–400	Local hardware
Sandbags (from §1 stock)	4	—	—
Pendulum + sweeper total		≈ 1,600–2,750	

Cut list

- Goalpost: 2 × 220 cm uprights, 1 × 220 cm top bar
- Sweeper bar (make 3 — 2 live + 1 spare, per needs doc §3): 1 × 100 cm foam bar core wrapped in noodle, 1 × 30 cm handle stub, each

Steps

1. Dry-fit the goalpost frame at the pinch gate — same foot-tee construction as the arch, sandbagged feet.
2. Tie the rope to the foam cylinder and hang it from the centre of the top bar so it swings clear across the gate.
3. Wrap pool-noodle lengths over a handle core to build each sweeper bar; cap both ends so nothing is exposed.
4. **Test:** swing the pendulum through its full arc — it must clear the goalpost uprights without snagging. Swing each sweeper bar low across the lane and confirm there's no rigid edge anywhere a car or a hand could catch (rulebook §7).

4. Spinner gate + Ramp Scramble

Spinner gate + ramp scramble

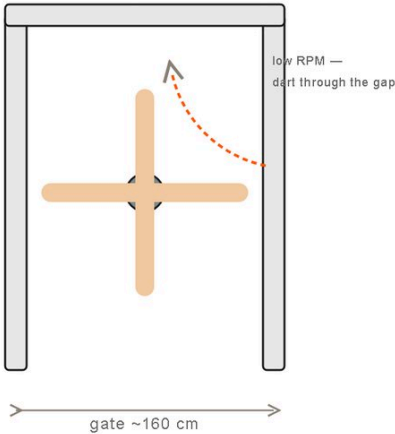
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Low-RPM motorised foam cross (needs doc §3) · wide low foam-topped ramp for the up-and-over round

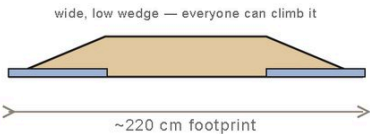
A · SPINNER GATE — 2 x 12 V gearmotor, PWM speed module

B · RAMP SCRAMBLE — ply core + EVA top, 1.2 m wide

gearmotor + PWM + SLA battery/PSU + printed hub, on the top bar



Bench-test the motor **BEFORE** the build weekend —
"the one true unknown in the kit" (Full Run Plan §1).
If it fails: Spinner Gate + The Grinder drop from the
shuffle, 19 of 21 rounds still run (Full Run Plan §1).
Foam arms only — cannot hit a person hard (rulebook §7).



Foam-topped so the crest stays soft AND grippy —
jams at the top are the content (Level Directory,
Tier-1 #7). Same build family as TorqBall's ramp kit.

The kit's one motorised obstacle, plus the wide low foam-topped ramp. **Bench-test the motors before the build weekend** — "the one true unknown in the kit" ([TorqFall — Full Run Plan §1](#)). If the bench test fails, Spinner Gate and The Grinder simply drop from the round shuffle — 19 of 21 rounds still run.

Materials

Item	Qty	Est ₹	Where
25 mm PVC pipe, 3 m (gate frame)	2 lengths	350–550	Local hardware
25 mm PVC elbows + tees	4	150–300	amazon.in
12 V low-RPM gearmotor (wiper-motor class)	2	1,600–3,200	amazon.in
PWM speed control module	2	600–1,200	amazon.in
SLA battery / PSU, 12 V	2	1,000–2,000	amazon.in
Printed hub (our 3D printer)	2	100–200 (filament)	—
Pool noodles (cross arms)	2	300–500	amazon.in
9 mm plywood, 8 × 4 ft (ramp core)	1 sheet	1,200–1,800	Timber yard
EVA foam sheet, 10 mm (ramp top)	1 m ²	500–1,000	amazon.in

Item	Qty	Est ₹	Where
Pine batten (ramp frame)	4 m	250–400	Timber yard
Spinner + ramp total		≈ 5,950–11,900	

Cut list

- Spinner gate: 2 × 220 cm uprights, 1 × 160 cm top bar; cross arms cut from noodle to ~70 cm each
- Ramp: 2 slope sides cut on the diagonal (full height to 0) over a ~220 cm total footprint; deck and EVA topsheet cut to match

Steps

1. **Bench-test** each gearmotor + PWM module + battery on the workbench first — confirm low, steady RPM with the multimeter before anything is mounted.
2. Dry-fit the gate frame, mount the tested motor at the centre with the printed hub, and zip-tie the foam cross arms on.
3. Build the ramp core from plywood and batten, cap the crest and both slopes with the EVA top so it's soft and grippy.
4. **Test:** run the spinner at low RPM and confirm a car-width gap stays open through the full rotation — foam arms only, it must not be able to hit a person hard (rulebook §7). Roll a car up the ramp; if it bottoms out at the crest, thin the frame or add EVA packing.

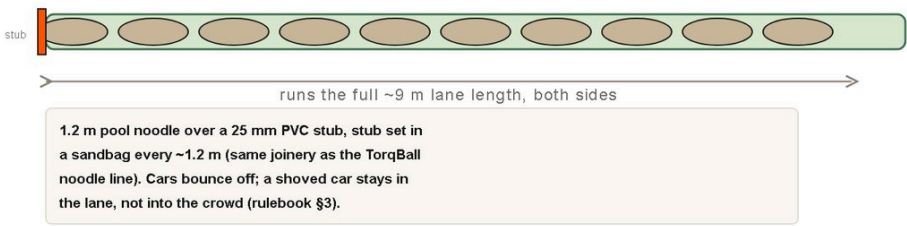
5. Soft barrier rail + bump rings + queue posts

Soft barrier rail, bump rings + queue posts

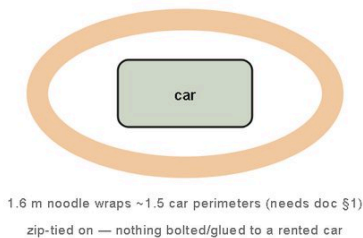
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Noodle-on-sandbag rail, full lane length, both sides (rulebook §3) — TorqFall's ONLY containment; no hard barricade anywhere

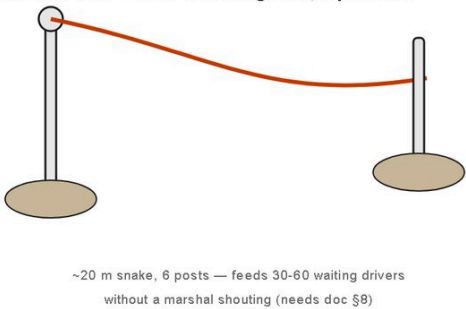
A · BARRIER RAIL — top view, one lane side



B · BUMP RING — per rented car, comes off clean



C · QUEUE POST — PVC + sandbag base, rope snake



TorqFall's **only** containment — there is no hard barricade in any scenario (rulebook §3). Noodle-on-sandbag rail runs the full ~9 m lane length, both sides; the same noodle stock also makes each rental car's bump ring and the queue-line posts.

Materials

Item	Qty	Est ₹	Where
Pool noodles, 1.2 m	~58–60 (barrier + gates + bump rings + spares, needs doc §3)	8,000–12,000	Wholesale/IndiaMART, amazon.in
25 mm PVC pipe, 3 m (stubs — barrier + queue posts)	~5 lengths	900–1,500	Local hardware
Sandbags (from §1 stock)	~40	—	—
Nylon rope, 20 m (queue snake)	1	300–500	amazon.in
Zip ties (from §1 stock)	—	—	—
Barrier + bump rings + queue total		≈ 9,300–14,200	

Cut list

- Barrier stubs: 25 mm PVC cut to 300 mm, one per noodle along both sides of the lane
- Queue posts (make 6): 1.4 m uprights, sandbag-cup base each

Steps

1. Cut PVC into 300 mm stubs. Slide a noodle onto each stub so the stub sits inside one end.
2. Lay noodles end to end along both sides of the lane, stub pushed into a sandbag every ~1.2 m.
3. Wrap a spare noodle length around each rental car's perimeter for the bump ring — zip-tied on, nothing bolted, screwed or glued to a rented car (needs doc §1; [Car Fleet, Scale & Protection](#)).
4. Build 6 queue posts (PVC upright in a sandbag base) and string rope between them in a snake for the queue line (needs doc §8).
5. **Test:** have a marshal shove a test car into the rail at heat speed — it should bounce back into the lane, never through the line.

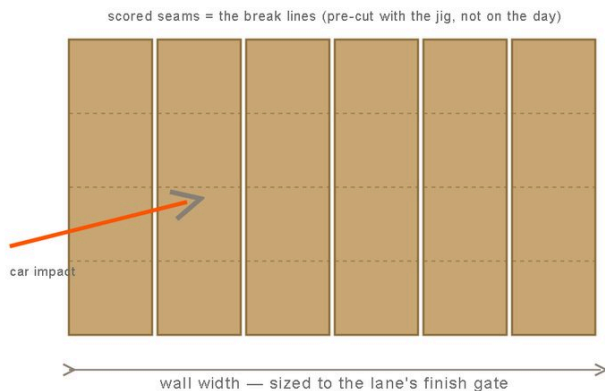
6. Collapse Gate wall

Collapse Gate — the smash finish

Pre-scored cardboard wall · every Grand-Final survivor punches through it (rulebook §2; Level Directory Tier-3 #6)

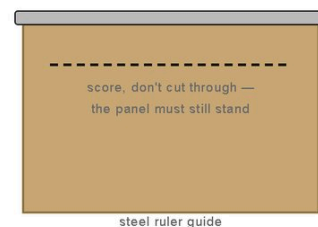
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A · WALL — scored panels, stacked into one breakable face



~40 panels/event (needs doc §3) — kabadiwala cardboard, same source + scoring jig as TorqSiege. Score seams with the shared cutting kit (knives, steel ruler, jig, mat — bought at T-13, priced in TorqSiege's needs doc, not re-priced here). Stack FRESH before the final (Full Run Plan, 17:10-17:30).

B · SCORING JIG — cuts the break-seams



TEST

Stack one wall, punch it at capped speed — it must fall dramatically, not just dent.
Second pass free if it doesn't break; re-score the seams live if it's still standing (Full Run Plan §3).

The Grand-Final smash finish — every survivor punches through a pre-scored wall to cross the line (rulebook §2; [TorqFall — Level Directory v1](#) Tier-3 #6). Same cardboard source and scoring jig as TorqSiege.

Materials

Item	Qty	Est ₹	Where
Cardboard sheets	~40 panels/event	500–1,000	Kabadiwala
Cutting kit — knives, steel ruler, scoring jig, cutting mat	1, one-time (shared with TorqSiege — bought at T-13 per TorqFall — Full Run Plan §1)	1,500	Local hardware / amazon.in
Collapse Gate total		≈ 2,000–2,500 first event · 500–1,000/event after	

Cut list

- Not cut to size — scored. Score each panel in a grid of horizontal seams so it still stands upright but breaks under a car hit; wall width is sized to the lane's finish gate.

Steps

1. Score every cardboard panel with the jig against the steel ruler guide — **score, don't cut through**; the panel must still stand on its own.
2. Stack ~40 scored panels into one wall at the finish gate, built **fresh before the final** ([TorqFall — Full Run Plan](#) §3, 17:10–17:30).
3. **Test:** stack one wall and punch it at capped speed — it must fall dramatically, not just dent. If it doesn't break, a second pass is free; if it's still standing, re-score the seams live.

7. Pit + charging table

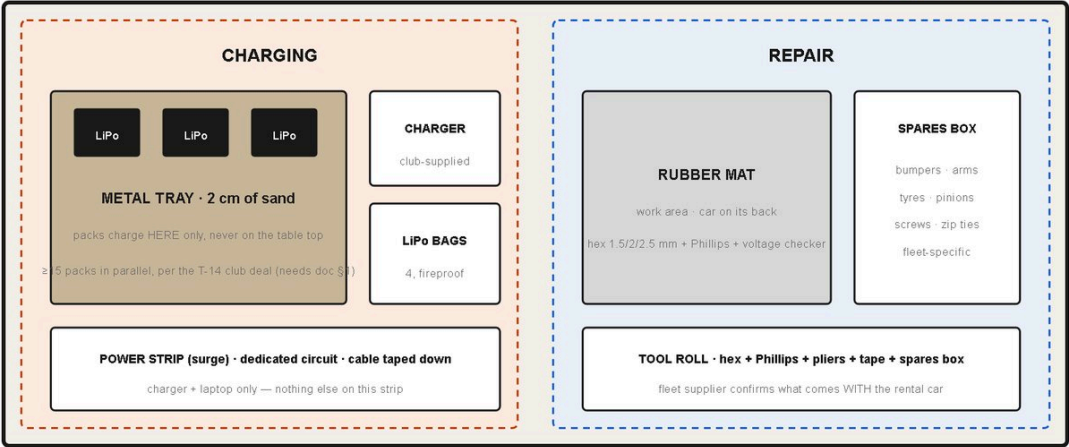
Pit + charging table — layout



One 6 ft folding table · charging on the LEFT, repair on the RIGHT · rental-fleet spares live here too (needs doc §7-8)

6 ft x 2.5 ft folding table (180 x 75 cm) — plan view

pit-cam (optional)



ON THE FLOOR, at the charging end



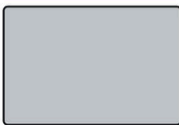
ABC extinguisher
4 kg



metal bin + lid
(puffed pack goes in)



bucket of sand



cement board / metal
sheet under the tray end

FLOOR under the charging end must be bare — never the turf, never a rented deck. One person = LiPo warden, first-aid lead nearby (needs doc §7-8; Event Ops Standards §5). Charge log every pack: # · in/out · voltage — the battery-endurance drill (Full Run Plan) needs this count every 15 min.

Mostly a **buy list**, not a build. The one rule that isn't negotiable: **LiPo packs charge on a metal tray of sand, on a bare floor, watched by one named person**. The rest of the protocol is in [Event Ops Standards \(all games\)](#) and [Car Fleet, Scale & Protection](#). TorqFall's fleet is **rented** — confirm with the RC club what charger ports, spares and tools come WITH the cars before buying anything twice (needs doc §1).

Item	Qty	Est ₹	Where
Folding table, 6 ft	1 (or borrow)	3,000–5,000	amazon.in
LiPo-safe charging bags	4	800–1,600	amazon.in
Balance charger, 2–4 port (only if the club doesn't bring enough — needs doc §1 needs ≥15 packs in parallel)	1	2,500–5,000	amazon.in
Steel tray, ~50 × 35 cm	1	400–800	amazon.in
ABC dry-powder extinguisher, 4 kg	1	1,800–2,800	amazon.in
Metal bin with lid	1	500–900	amazon.in
Surge-protected power strip, 6 socket, 3 m lead	1	800–1,500	amazon.in

Item	Qty	Est ₹	Where
Rubber work mat, 60 × 40 cm	1	400–800	amazon.in
RC hex driver set 1.5 / 2.0 / 2.5 + Phillips	1	500–1,200	amazon.in
LiPo voltage checker / cell meter	1	300–600	amazon.in
Cement board or steel sheet, ~1 m ² (floor under the tray end)	1	400–800	Local hardware
Sand (from §1 stock), clipboard, marker	—	100–200	—
Fleet-specific spares box (confirm with the RC club first — needs doc §1)	1	1,500–3,000	Fleet supplier / amazon.in
Pit total		≈ 12,700– 23,900	

Setup (1 hour, on the day)

1. Table goes on **bare floor** — pull the turf back or lay the cement board under the charging end. Never on turf, never on the rented driver deck.
2. Charging end: tray with 2 cm of sand → packs lie on the sand while charging → charger beside it (club-supplied or ours) → LiPo bags for anything not charging. Extinguisher, sand bucket and metal bin on the floor at that end.
3. One power strip, one wall socket, cable gaffer-taped to the floor — charger + laptop only on that strip.
4. Repair end: rubber mat, spares box, tool roll.
5. Name the **LiPo warden** at the crew brief — the day's first-aid lead should be nearby (needs doc §7–8). They log every pack in/out with voltage; the battery-endurance drill needs this count every 15 minutes ([TorqFall — Full Run Plan](#) §1).

8. PVC + plywood cut plan

Take this to the pipe/timber yard on D5 morning — most Mumbai pipe and ply shops cut to size for ₹20–40 a cut.

Use	Length needed	Section
Start/finish arches (2)	~10 m	§2
Pendulum goalpost	~9 m	§3 (needs doc §3)
Spinner gate frame	~4 m	§4 (needs doc §3)
Sweeper handles	~3 m	§3 (needs doc §3)
Rising-gate / T-feet stock (kit-wide, other Tier-2/3 rounds)	~10 m	needs doc §3
Barrier stubs, both sides of the lane	~15 m	§5
Queue posts (6)	~8 m	§5

Use	Length needed	Section
Spare	~1 m	—
Total ≈ 60 m 1" PVC		matches TorqFall — What The Event Needs §3

Sheet	Thickness	Yields
1 × 8×4 ft sheet	9 mm	Ramp Scramble core (§4) — the only plywood item in the kit

9. Build order — D5 to D8, 2–3 people

Slots into the [TorqFall — Venue Scenarios & Build Plan v1](#) §5 timeline.

Day	Morning	Afternoon	Needs
D5	Pipe/hardware run with the §8 cut plan	Cut PVC, dry-fit both arches + pendulum goalpost + spinner frame	Hand saw, tape measure
D6	LED strips on arches; pendulum + sweeper bars assembled	Spinner motor bench test (the one true unknown) + ramp build	Multimeter, hot glue
D7	Barrier rail laid out on both sides + queue posts built	Collapse–Gate panels scored with the jig	Jig, steel ruler, cutting mat
D8	Dry-fit everything in the garage as one layout, fix what's wrong	Pack into labelled crates, charge power banks + spinner batteries	—

D10 is the full rehearsal — everything built here goes up once, at full size, before the venue.

10. Cost roll-up

TorqFall's DIY kit doesn't change by venue scenario — there's no hard barricade to add in Scenario A and no soft-line-only discount in B/C, since the barrier rail is identical everywhere (rulebook §3). One total covers all three:

Item	Est ₹
Start/finish arches (§2)	3,700–6,900
Pendulum + sweeper bars (§3)	1,600–2,750
Spinner gate + ramp (§4)	5,950–11,900
Barrier rail + bump rings + queue posts (§5)	9,300–14,200
Collapse Gate wall + jig, first event (§6)	2,000–2,500
Pit + charging (§7)	12,700–23,900
DIY build total	≈ 35,250–62,150

Item	Est ₹
Tools if none owned (§1)	+6,500–10,500

This reconciles with [TorqFall — What The Event Needs](#) §3's gauntlet-kit subtotal (₹40,000–61,000) plus its §7–8 pit/ops lines — the two documents price the same physical build from different angles: this guide itemises the shopping list and steps, the needs doc prices the whole event. Vendor lines (turf, driver stand, branding print, DJ/sound, food, cars + pit crew, and Scenario C's tent + AC) sit on top — see the Venue Scenarios doc §4.

11. Safety while building

- The **soft-only design law** applies to everything built here — no rigid edge, no trap, no flip-launch, resets in under 3 seconds (rulebook §3, §7).
- Sand every cut edge before it goes near a car or a hand — splinters snag a chassis and cut a marshal's palm.
- **Bench-test the spinner motors before mounting** — foam arms only, low RPM, cannot hit a person hard.
- Nothing electrical beyond USB LED strips and the low-voltage spinner motors is DIY here; mains stays with the venue's dedicated circuit.
- LiPo protocol is not a build step — it's a person. Name the warden before anything charges, and keep the first-aid lead nearby (needs doc §7–8; Event Ops Standards §5).
- Zip-tie, never bolt or glue, anything that touches a rented car (needs doc §1; [Car Fleet, Scale & Protection](#)).